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AI-based wearable vision assistance system for the visually impaired: Integrating real-time object recognition and contextual understanding using large vision-language models

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ABSTRACT

Visual impairment affects the ability of people to live a normal life and perform daily life activities. Traditional approaches to helping visually impaired people are deficient in obtaining contextually rich environmental information. This research introduces a novel large vision language model (LVLM) based wearable vision assistance system with a hat-mounted camera connected to a Raspberry Pi 4 Model B to recognize new people or objects through a one-click process. The system allows users to add data on new individuals and objects for later recognition, enhancing the accuracy of the recognition over time. LVLM enables richer environmental descriptions compared to traditional CNN-based systems. The proposed system achieved 90% object recognition accuracy in well-lit conditions and 85% average usability score from users participated in usability study, indicating high reliability of the system and users satisfaction. Additionally, the proposed system incorporates a distance sensor that activates a beeping sound using a buzzer as soon as the user is about to collide with an object, helping to ensure safety while navigating their environment. A comprehensive comparative study is carried out to evaluate the proposed system against traditional support techniques. Comparative analysis shows that the proposed system demonstrates a promising approach for integrating contextual understanding and personalization in wearable assistive technologies and solves the major issues faced by the community of visually impaired people. This solution has practical use in environments such as smart shopping, urban navigation, and public transportation systems, making it adaptable to both personal and smart city contexts.

1. Introduction

Globally, more than 7.5 million people suffer from moderate to severe forms of visual impairment [1]. People with limited or no vision face challenges related to mobility, independence, and social engagement [2–4], necessitating innovative solutions that transcend conventional aids.

Conventional assistive technologies, while invaluable, possess inherent limitations [5]. White canes, for example, primarily offer tactile feedback for immediate obstacles at ground level, but do not convey information about signage, distant hazards, or specific object identities [6]. Guide dogs, while providing exceptional mobility support, require extensive training and care, and their understanding is not based on visual semantic interpretation [7]. Sensor-based navigation aids [8] often provide simplistic proximity alerts, however, lack in adaptability, personalization, and advanced features [9]. Furthermore, power consumption of these devices is also an issue when used in outdoor environments [10,11]. Such limitations prompted the development of AI-based solutions to provide improved functionality and real-time adaptability.

AI based advanced solutions have opened up new possibilities for visually impaired people and have improved their overall quality of life. Systems such as smart assistive sticks [12], Smart belts, and Glasses [13,14], while using AI and computer vision algorithms and real-time positioning systems are capable of providing more accurate

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and effective navigation solutions with real-time feedback [15–17]. These devices integrate a multitude of advanced functions while maintaining a lightweight, low-power, and highly efficient design, offering users optimal comfort and performance [18]. Recent AI solutions such as SeeingAI [19], NoorCam MyEye [20], and Be My Eyes [21] can interpret visual data, identify objects, faces, text, scenes, and even emotions, and then communicate this information through auditory or haptic feedback [22,23]. In this way, based on the achieved information, a detailed guided response can be obtained.

Although, there exist numerous AI based solutions [24–26] They often lack customization, flexibility, and contextual awareness required to support users in dynamic environments [24,27–29]. Most existing systems focus primarily on static object detection and limited scene interpretation, with insufficient contextual reasoning and limited capability to adapt to individual user needs or environments. Additionally, most commercial solutions rely on fixed datasets and do not support dynamic addition of new objects or faces without retraining. Hence, advanced assistive technologies are in increasing demand that can allow visually impaired people to live independently and have a thorough awareness of their surroundings [30,31]. This technological gap underscores a persistent demand for systems capable of delivering comprehensive and nuanced environmental information.

This study presents a LVLM based wearable vision assistance system with a hat-mounted camera connected to a Raspberry Pi 4 Model B. Using a hat-mounted camera, the system recognizes real-time objects and LVLM provides a rich contextual understanding of the environment.

The main contributions of this work are:

- Development of a wearable assistive vision system integrating LVLM for contextual scene understanding.
- Implementation of real-time object recognition using MobileNetV2 optimized for wearable hardware.
- Introduction of a personalized object and face recognition module using embedding-based matching with FaceNet.
- Design of a user-adaptive learning mechanism enabling dynamic addition of objects and faces without full retraining.
  Experimental evaluation demonstrating practical usability for visually impaired assistance.

The novelty of the proposed system lies in the integration of three complementary capabilities within a unified edge–cloud architecture designed for wearable deployment. First, the system performs real-time object detection directly on-device using a lightweight MobileNetV2-SSD model optimized through TensorFlow Lite, enabling continuous environmental perception on resource-constrained hardware such as the Raspberry Pi 4. Second, the system introduces a personalized recognition mechanism that allows users to incrementally add new faces or objects through a one-click capture and voice-labeling workflow. Instead of requiring model retraining, the system stores feature embeddings generated by FaceNet in a local database and performs similarity matching during inference. This enables efficient user-specific recognition while maintaining real-time performance.

Unlike existing commercial assistive systems such as Seeing AI [19] and NoorCam MyEye [20], which primarily provide predefined recognition capabilities or rely heavily on cloud services, the proposed approach combines edge-based detection, incremental personalization, and LVLM-based contextual reasoning. The integration of a large vision-language model allows the system to generate relationship-aware scene descriptions and contextual explanations, rather than simple object labels. This hybrid architecture demonstrates that context-aware assistive perception and user-driven personalization can be achieved on affordable wearable hardware, making the approach both technically and practically distinct from existing solutions.

For validating the presented system an extensive user testing was carried out with participants who have visual impairments to assess the system's efficiency. The findings indicate a notable improvement in user environmental awareness, social engagement, and autonomy.

The remainder of the paper is organized as follows.

Section 2 presents the latest literature related to the presented study. Section 3 provides the details of the system design and development including benchmarking datasets used, object and face detection, integration with real-time object detection and Large Language Models (LLM), etc. Section 4 presents the details of the evaluation methods used to evaluate the presented system. Section 5 presents the results achieved by the system and discussion on them. Section 6 provides detail of system's practical applications and possible extensions. Section 7 provides challenges and limitations of the presented system. Section 8 presents the future directions and improvements and finally, Section 9 provides the conclusion of the study.

2. Literature review

2.1. Conventional assistive technologies

Different assistive technologies have been presented in the literature to help visually impaired individuals achieve greater independence. Among them, white canes remain one of the most widely used mobility aids because they provide tactile feedback that allows users to detect obstacles [6,32,33]. However, due to a number of their shortcomings, including their inability to identify overhanging objects, head-level impediments, and trunks. Additionally, their insistence on making direct physical contact with the obstacles and their failure to provide guidance on how to approach them caused researchers to focus on various alternative technologies and approaches [32].

Assistant dogs have been utilized as an alternative to white canes to overcome their limitations [34,35]. These dogs provide both navigation supports and companionship. However, obtaining and maintaining an assistant dog requires extensive training, time, and financial resources [36–38]. This leads to replace assistant dogs with robotic dogs as suggested in different studies [39–42]. Robotic dogs present a great deal in improving the quality of life of visually impaired people, however, developing robust and effective robotic mobility aids with static and dynamic settings for use in both indoor and outdoor environments has remained a difficult task to accomplish [43].

2.2. Sensor-based assistive technologies

Advances in sensing technologies have led to the development of sensor-based assistive devices such as smart canes, ultrasonic obstacle detectors, and GPS-based navigation systems. These devices detect obstacles and provide feedback through auditory or vibration cues to assist users during navigation [44]. While these systems improve obstacle detection and basic navigation, they generally lack the ability to interpret complex environments or provide contextual information. As a result, visually impaired individuals may still face difficulties in performing daily activities that require environmental understanding, such as shopping or locating specific items.

2.3. Wearable assistive technologies

Important progress was made with the development of wearable devices such as eSight [45] and NoorCam MyEye [20]. These devices offer visual enhancement and optical character recognition (OCR) capabilities [46]. Chest-mounted cameras are used with computer vision systems installed in them [47]. These devices are linked with smartphones for processing and connectivity purposes [48]. Despite these great works, many wearable systems still struggle to provide meaningful contextual descriptions of complex scenes [49], limiting their ability to support independent interaction with dynamic environments.

Table 1
Comparative summary of AI powered assistive technologies and the proposed system.

System	Key features	Limitations
NoorCam MyEye [20]	Text reading (printed and digital), Face recognition, Natural language understanding, Light weight	Limited (manual updates), Expensive, minimal contextual info, Works well in good lighting
Seeing AI [19]	Text recognition (printed and handwriting), Scene Description, Face recognition Photo descriptions,	Moderate (Scene Describer), Cloud dependent, latency issues
Be My Eyes [21]	It connects blind or low-vision users with human volunteers through live video calls.	Requires volunteers & network
NavCog [53]	Smartphone-based navigation app (developed by Carnegie Mellon); provides turn-by-turn navigation using GPS, Bluetooth beacons, and audio feedback.	Limited contextual understanding
Sunu Band [54]	Smartwatch-style wearable with sonar/ultrasonic sensors for obstacle detection; provides haptic feedback (vibrations) to indicate proximity of objects; includes navigation features when paired with a smartphone.	Limited scene/contextual description (focuses only on obstacle distance)
Proposed System	(MobileNetV2 SSD + FaceNet), (user can add new faces/objects), (user can get deep description of the image)	Dependent on cloud for deep contextual queries

2.4. AI based assistive technologies

Recent advances in artificial intelligence have significantly improved assistive technologies for visually impaired individuals. AI-based systems can analyze visual data using computer vision algorithms to detect objects, recognize faces, read text, and interpret scenes [50, 51]. Several mobile applications, including Seeing AI [19] and NoorCam MyEye [20], use these capabilities to assist users in recognizing objects and reading text in real time [52]. Although these systems offer notable improvements, challenges remain in developing affordable, personalized, and context-aware solutions that can fully support independent daily activities.

2.5. Summary of the existing systems and the proposed approach

Table 1 summarizes existing AI-based assistive technologies and compares them with the proposed system. While current solutions provide various functionalities, a common limitation is their inability to deliver detailed contextual descriptions of the surrounding environment. In addition, most systems do not allow users to add new people or objects to a personalized recognition database, which limits their adaptability to user-specific environments.

To address these limitations, the proposed system introduces a Large Language Model (LLM)-based framework that generates richer contextual descriptions of the user's surroundings (see Figs. 8 and 9). The system also enables users to add new faces and objects to a personalized database through a voice-activated capture process, allowing the dataset to expand progressively and improving recognition accuracy over time.

Unlike existing assistive systems, the proposed approach integrates three key components within a unified architecture: lightweight on-device detection, incremental personalized recognition, and LVLM-based contextual reasoning. This design enables both object recognition and relationship-aware scene understanding while remaining suitable for deployment on resource-constrained wearable hardware.

Furthermore, the system supports incremental personalization through a one-click capture and voice-labeling mechanism. Instead of retraining the model, feature embeddings and labels are stored in a local SQLite database and matched during inference. This approach allows efficient personalization on devices such as the Raspberry Pi 4 while maintaining real-time performance.

3. System design and development

3.1. System design

The design of the proposed system is based on three operational systems as shown in Fig. 1.

One system serves as the core (if no button is pressed), continuously running, while the other two are activated upon button press. There are two buttons, a green button and a blue button. If no button is pressed, the system (MobileNetV2 SSD) automatically detects objects (only nearby and ignore the far away). If the detected object is already in the system's database, the system conveys its information to user through voice using text-speech processing module i.e., Festival (refer to Fig. 3). If the detected object is not in the system's database, the system gets trained on it and save this entry into the SQLite database for later detection. The user presses green button when he wants to add a new object or person in the database for the later recognition (refer to Fig. 1).

When the green button is pressed, the system captures image of an object (for which user wants). User then speaks the name of the object, which is converted to a text through TensorFlow speech to text module. The TensorFlow module auto labels the object by the user's spoken name. The FaceNet model is fine-tuned on this new labeled object and the SQLite database gets updated on it for later recognition. During training, the system extracts feature embeddings from the captured face or object, which, along with the user-provided label are stored in a local SQLite database. During subsequent recognition, the newly captured embeddings are compared with stored embeddings using cosine similarity to determine the closest match. This process is outlined in Figs. 5, 6, and 7.

Pressing the blue button directs the system to capture an image, which is then processed by the LVLM model to generate detailed information about the image. This detailed information of the image is conveyed to the user by the audio output generated by the text-speech system i.e. Festival. This process is shown in Figs. 8 and 9. After each step, the system waits for the next action or continues running until it is turned off.

The system does not perform full retraining of MobileNetV2 SSD or FaceNet during runtime due to edge hardware constraints. Instead, when a user adds a new object/face (green-button workflow), the system extracts feature embeddings and stores them with the user-provided label in a local SQLite database. During subsequent operation, recognition is performed via embedding comparison (cosine similarity) against the stored database entries. MobileNetV2 SSD weights remain fixed during runtime; personalization is achieved through incremental database expansion rather than continuous weight updates.

3.2. Hardware components

The following hardware components have been used in the proposed system design (also shown in Fig. 2).

Raspberry Pi 4 Model B (8 GB RAM): The processing unit of the system chosen for its balance of performance and energy efficiency. It is capable of running AI models with great speed.

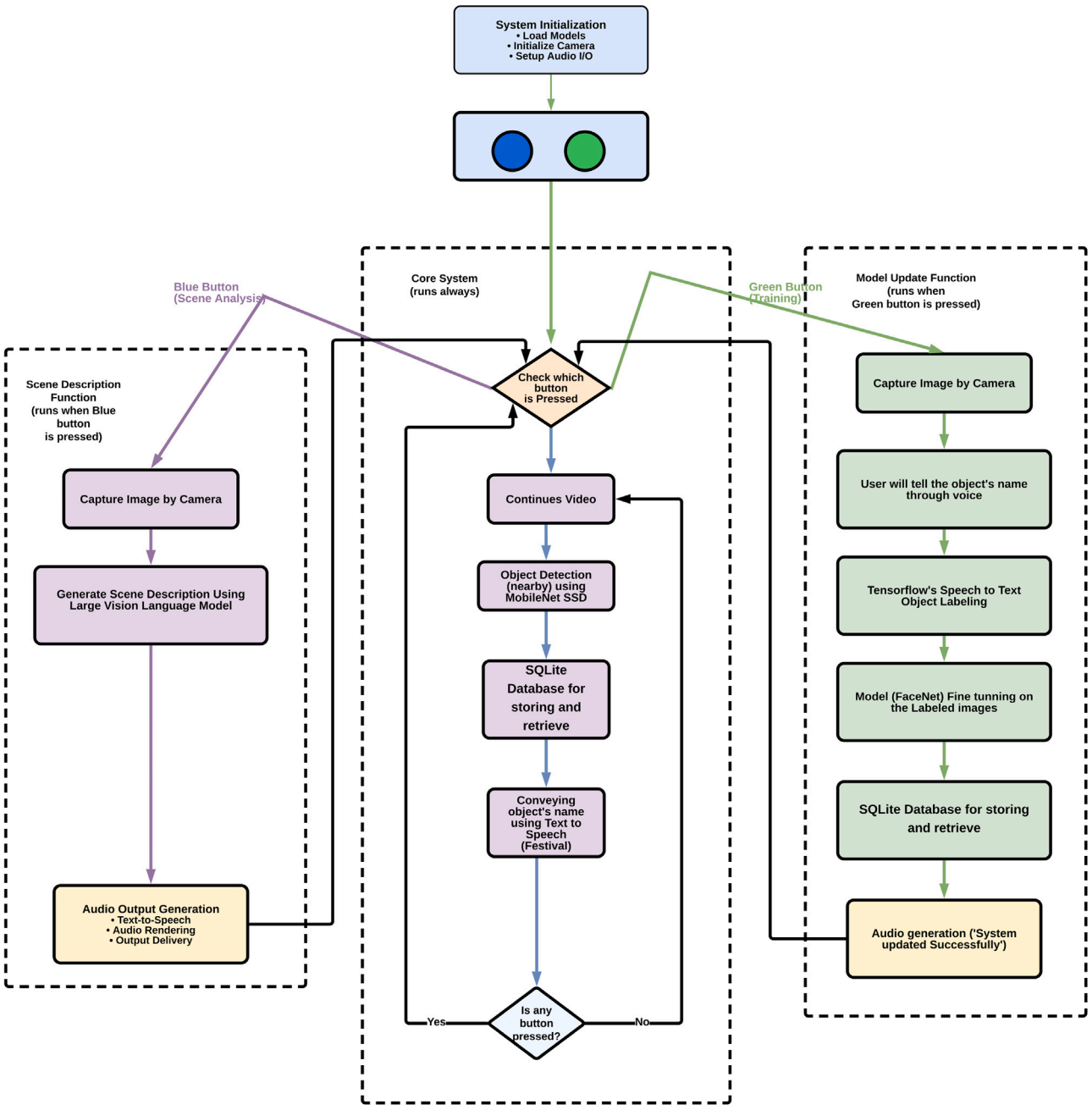


Fig. 1. Proposed system design.

Camera Module: An 8MP Camera Module for Raspberry Pi, built-in to the brim of a hat, that is lightweight, and high-resolution to capture photos from the subject's point of view.
Camera Support Hat: A hat that is specially designed to enable the installation of a camera.
Power Supply: A rechargeable battery with a capacity of 20,000 mAh is stored in a carrying case.
Audio Output: Stereo-acoustic headphones allow sound to travel directly through the skull, so the user can hear clearly without covering the ears. The user can also hear outside of this device to stay alert about the surroundings [55].
System Distance Calculation: An ultrasonic sensor helps to easily determine how far away an object is by using sound waves, making it simple to determine distances.
Buzzer: A buzzer is triggered to warn the user and help avoid collisions when an object is less than 20 cm.

User Interface Button:

- **Add Item Buttons**: A physical press button (green button) located on the hat enables users to input new people or objects into the system with their labels.
- **Button for Contextual Details**: Another physical press button (the blue button) prompts the system to deliver a detailed description of the currently focused object using LLM.

3.3. Software architecture

The software architecture of the proposed system includes several modules designed for efficient data processing and user interaction as shown in Fig. 3.

The Raspberry Pi runs on a Linux-based operating system (Raspbian) with real-time processing capabilities. The image acquisition

Table 2
Technical implementation details of the proposed system.

Component/Metric	Specification/Value	Notes
Processing Unit	Raspberry Pi 4 Model B (8GB RAM, ARM Cortex-A72 1.5 GHz)	Balanced performance/Energy efficiency
Camera Module	8MP Raspberry Pi Camera Module	Mounted on wearable hat
Battery	20,000 mAh rechargeable power bank	~6–8 h continuous use
Audio Output	Bone-conduction stereo headphones	Keeps ears open for environment sounds
Models Used	MobileNetV2 SSD (object detection, quantized TensorFlow Lite); FaceNet (face recognition); GPT-4o mini (contextual understanding via Azure)	Optimized for ARM
(Local)	~1.5 s per frame (MobileNetV2 + FaceNet)	Includes preprocessing
Contextual Query Latency	~5.5 s (due to GPT-4o mini API call)	Network-dependent
Energy Consumption	~4.5 W average (camera + Pi + TTS)	Battery drains ~20%/h
Thermal Constraints	Mild throttling after >3 h continuous use	Mitigated by passive cooling case



Fig. 2. Hardware components of the proposed system.

module uses OpenCV [56] to capture images from the camera, manage frame rates, and preprocess images. The system implements MobileNetV2 SSD [57] for object detection and FaceNet [58] for face recognition, both optimized for the Raspberry Pi's ARM architecture using TensorFlow Lite [59].

To improve performance, models are quantized to reduce their size and improve inference speed without sacrificing significant accuracy [60]. A lightweight database, SQLite, is used to manage the feature embeddings and associated labels to enable quick retrieval during recognition tasks. The system integrates a large language model through API calls to OpenAI's GPT-4o mini via Azure. The audio processing module utilizes local-speech systems, such as Festival [61], to convert written text into speech. User inputs are managed through the user interface module, which provides feedback and guides the user through interactions.

Table 2 provides technical implementation details of the system such as model size, hardware specifications (e.g., camera-to-response latency), and energy consumption of the system.

3.4. Bench-marking datasets used

The proposed system was trained on a combination of publicly available benchmarking datasets and a real-world dataset collected during usability study. The following datasets were utilized in the development and assessment of the system.

COCO Dataset [62]: Pre-trained weights from the Common Objects in Context (COCO) dataset were employed for the MobileNetV2 SSD object detection module. Transfer Learning (TL) was applied by fine-tuning the classification layers using a subset of images captured during system pilot testing to improve recognition of locally relevant objects.
Labeled Faces in the Wild (LFW) Dataset [63]: The FaceNet model utilized embeddings generated from the LFW dataset for initial face recognition capability. Additional real-world images were incorporated incrementally during the usability study to enhance system personalization and performance.
Flickr8k Dataset [64]: This dataset was exclusively used for evaluating the contextual description module integrated with GPT-4o mini. Generated descriptions were benchmarked against ground-truth captions using the BLEU score metric.

3.5. Data acquisition and processing pipeline

The data acquisition and processing pipeline of the proposed system consists of the following sequential steps.
Image Acquisition: The system employs an 8MP Raspberry Pi Camera Module mounted on the brim of a wearable hat to continuously capture images from the user's point of view. Image capture occurs in two modes:
Automatic Scanning Mode: The core system runs continuously, capturing frames at regular intervals for obstacle and face detection.
On-Demand Mode: When the user presses the blue or green button, the system captures a snapshot for contextual understanding or personalized recognition respectively.
Preprocessing: Captured images are resized to 300×300 *and* 160×160 *pixels* to meet the input specifications of the MobileNetV2 SSD and FaceNet models respectively. Basic preprocessing operations such as resizing, normalization, and format conversion are performed using the OpenCV library.

3.6. Object and face detection

Object Detection: The preprocessed image is passed to a *MobileNetV2 SSD* model, where TL was employed using newly captured local images. Due to hardware constraints, full cross-validation was not applied. Instead, a hold-out test set strategy was used for performance validation. Detected objects are filtered using a confidence threshold (greater than 50%) and Non-Maximum Suppression (NMS) is applied to remove redundant detections.
Face Detection and Recognition: Faces are detected using a *Haar cascade classifier* [65]. Detected faces are converted into *128-dimensional embeddings* via *FaceNet* model. These embeddings are compared with the existing entries in the personalized SQLite database using *cosine similarity* for recognition.

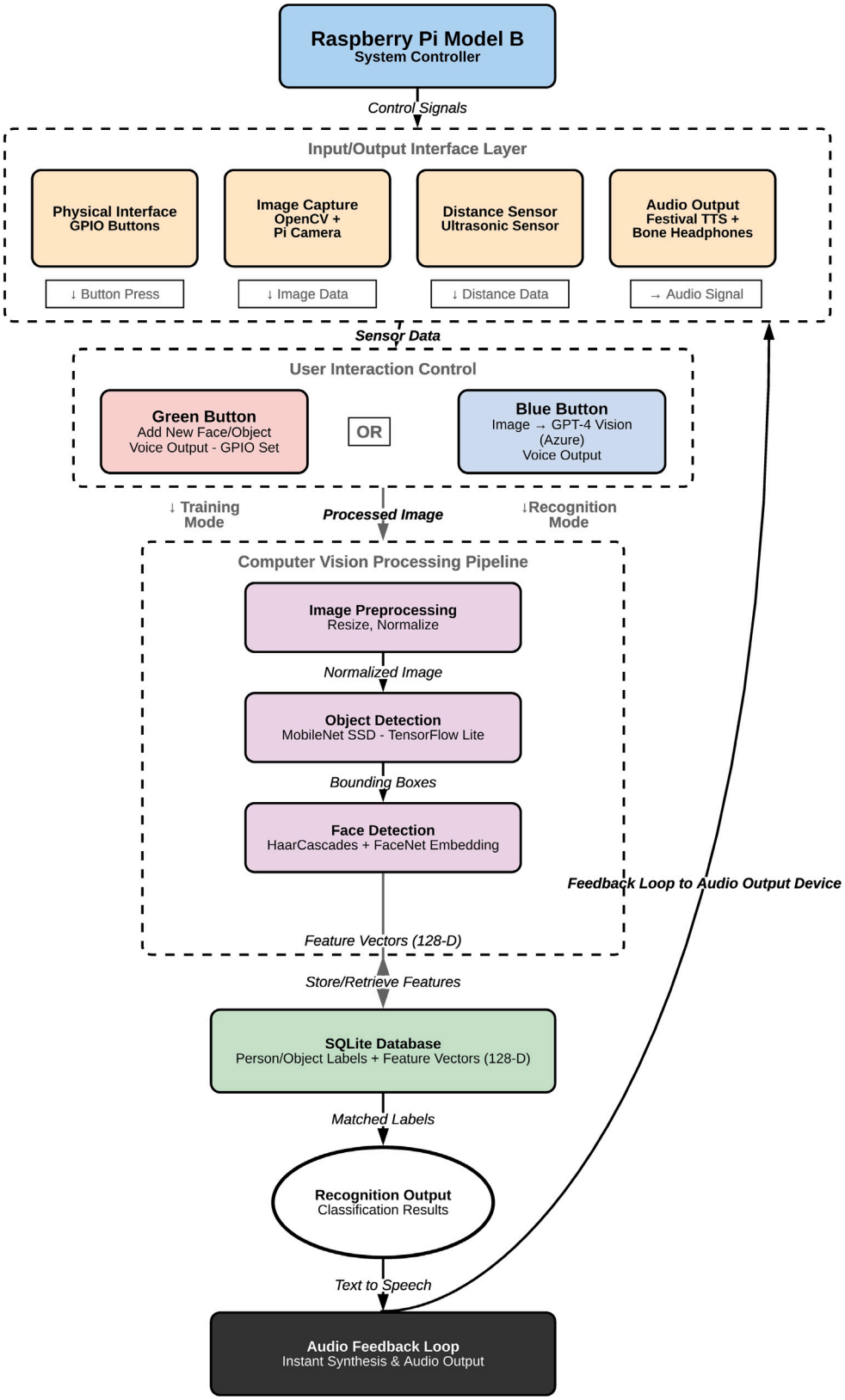


Fig. 3. Software architecture of the proposed system.

3.7. Integration with real-time object addition (Green button workflow)

Due to the computational limitations of embedded hardware, the system does not perform full retraining of MobileNetV2-SSD or FaceNet during runtime. Instead, personalization is implemented using an embedding-based incremental recognition strategy. When a user registers a new object or face using the green-button workflow, the system captures an image and extracts feature embeddings using the pre-trained model. These embeddings, together with the user-provided label obtained via voice input, are stored in a local SQLite database.

During subsequent operation, embeddings extracted from newly captured images are compared with stored entries using cosine similarity to identify the closest match. The underlying MobileNetV2-SSD and FaceNet model weights remain unchanged during runtime. This approach enables real-time personalization without the computational cost of retraining deep neural networks on resource-constrained hardware such as the Raspberry Pi.

3.8. Integration with large language models

When the blue button is pressed, the system initiates a sequence of actions. It captures an image of an object and then by identifying its label and spatial relationships, changes it to a descriptive prompt. An API call is made, where the generated prompt is sent to the OpenAI's GPT-4o mini model via Azure for generating rich contextual descriptions of the environment. The resulting text includes object relationships, environment cues, and suggestions for user interaction. The response handling step involves processing the output from the model and formatting it to be suitable for speech conversion.

The captured image is first processed by the object detection module based on MobileNetV2. Detected objects are then passed to the Large Vision–Language Model (LVLM), which generates contextual descriptions of the surrounding environment. This multimodal reasoning enables the system to interpret relationships between objects and produce meaningful scene-level descriptions rather than isolated labels.

3.9. Parameters tuning

For object detection, the MobileNetV2 SSD model was fine-tuned using a subset of locally captured images. Key hyperparameters such as learning rate (set to 0.001), batch size (32), and optimizer (Adam) were chosen based on standard practices and empirical performance on the hold-out validation set. Training was conducted over 10 epochs, with early stopping not applied due to limited data and the rapid convergence observed in loss curves (see Fig. 12).

For LVLM integration, GPT-4o mini was used in zero-shot mode via Azure API. The system-generated prompts were sent using a fixed temperature of 0.7 and a maximum token limit of 150. These values were chosen to balance response quality and latency during real-time operation. FaceNet was used with default embedding parameters without additional retraining. It is important to note that the MobileNetV2 and FaceNet models are not retrained during system operation. Instead, newly introduced objects or faces are represented using extracted feature embeddings that are stored in a user-specific database. Recognition is performed through similarity matching between live embeddings and stored vectors, enabling real-time personalization without modifying network weights.

3.10. Audio feedback generation

The final output, either brief detection alerts or detailed contextual descriptions, is converted to spoken auditory feedback using a *local-speech engine (Festival)*. Audio output is transmitted via *bone conduction headphones*, allowing users to hear system responses while maintaining environmental awareness.

3.11. User interaction and database update

Users can add new objects or faces to the personalized database by pressing the *green button*, providing a label via voice input, which is transcribed and stored alongside the captured image's features. This enables continuous system learning and improves recognition accuracy over time.

3.12. Collision detection and alert system

An ultrasonic sensor continuously measures the distance to nearby obstacles. If an object is detected within *20 cm*, a *buzzer is activated* to alert the user, enhancing safety during navigation.

3.13. Security and protection

There is a strong security system that guarantees the protection of the information provided by the users. The data minimization principles are adhered so that only required information is transmitted; in case of buttons being used by the user to start a process, images are not transmitted to OpenAI unless initiated by a button. All API calls to OpenAI are made securely through Microsoft Azure; the call itself is encrypted to increase the level of security for users.

3.14. Privacy and ethical concerns

The use of facial recognition technology, particularly in assistive devices for vulnerable populations such as visually impaired individuals, necessitates careful attention to ethical and privacy concerns. The proposed system requires explicit user initiation (via physical button press) for both face addition and contextual image analysis, ensuring that image capture is under user control. Facial data, including embeddings and labels, are stored locally on the device using an encrypted SQLite database, and no data is transmitted to cloud servers unless the user explicitly activates contextual description features.

4. System evaluation

4.1. Evaluation based on performance metrics

The performance of the proposed system has been evaluated using several important evaluation parameters that include Precision, Recall, F-Score, and Accuracy values. Each of these parameter is briefly explained below.

Precision: Precision refers to the proportion of the predicted positives of the model that are positive. Mathematically,

$$Precision = \frac{TP}{TP + FP} \tag{1}$$

Recall: Recall refers to the proportion of the actual positives correctly classified as positives by the model. Mathematically,

$$Recall = \frac{TP}{TP + FN} \tag{2}$$

F-Score: F-Score is the harmonic mean of the Precision and Recall. Mathematically,

$$F - Score = \frac{2 * TP}{2 * TP + FP + FN} \tag{3}$$

Accuracy: Accuracy refers to the proportion of all the correct classifications of the model, Mathematically,

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{4}$$

In all of the above equations, TP, TN, FP, and FN are the values taken from the confusion matrix. Where;

TP (True Positives): represents the actual positives predicted as positive by the model,

Table 3
Demographic breakdown of the participants of the usability study.

Category	Subcategory	Count	Percentage (%)
Gender	Male	25	50%
	Female	25	50%
Age Group	18–30	15	30%
	31–50	20	40%
	51–70	15	30%
Visual Impairment Level	Low Vision	30	60%
	Blind	20	40%

TN (True Negatives): represents the actual negatives predicted as negative by the model,
FP (False Positives): represents the predicted positives by the model, which are negatives,
FN (False Negatives): represents the predicted negatives by the model, which are positives.

4.2. Evaluation based on usability study

For the usability evaluation of the proposed system, 50 visually impaired participants having 25 men and 25 women, aged between 18 and 70 years, have been selected. Table 3 provides more information about the participants of the usability study, their vision impairment levels and their age groups.

As seen from Table 3 that participants with different visual loss levels have been selected to make the study diverse and provide a wide range of experience.

The participants tested the device in conditions that were created to resemble real-life situations. We created situations such as moving inside, identifying objects in a kitchen, having conversations in a simulated cafe, and even picking up fruits they want to eat. This arrangement enabled us to determine how well the device would function in real-life scenarios.

5. Results and discussion

5.1. Recognition accuracy, precision, recall, and F-score values

The results achieved by the proposed system are summarized in Table 4 and illustrated by the confusion matrix in Fig. 4. As shown in Table 4, under good lighting conditions, both the object recognition module and the contextual LVLM system achieved an accuracy of 90%. This level of performance is often challenging for traditional Deep Learning (DL) models in the context of visual understanding. This is attributed to the fact that LLM are trained on vast amounts of textual data, enabling them to generate richer and more context-aware descriptions compared to traditional DL models [66,67]. Furthermore, unique architecture and training approach of LVLM make them capable to exhibit human-like responses to images by integrating visual and textual processing abilities [68]. Table 4 also shows that in the good lighting situations, the system achieved desirable values of Precision, Recall, and F-measures, indicating the good performance of the system. In low lighting conditions, the accuracy dipped to 80%. In this case, both precision and recall were at 80%, indicating that 80% of the objects detected by the system were correctly identified, while 20% were falsely recognized. The confusion matrix presented in Fig. 4 shows that 90% of the objects detected by the proposed system are identified correctly, however, there were still some mistakes i.e. 10% of the objects are detected as false positives. This indicates that the system incorrectly identified an object that was not present.

Table 5 further provides details about object recognition by category. The system achieved the highest score for navigation obstacles (F1 score: 0.93) and food items (F1-score: 0.93), which are critical for safety and daily independence. Furniture recognition showed slightly lower performance (F1 score: 0.86) due to ambiguous shapes and visual similarities between different types of furniture. These category-specific insights guided our user study design, ensuring comprehensive testing across diverse real-world scenarios.

Table 4
Precision, recall, f1-score, and accuracy for good lighting and low lighting conditions of overall system.

Condition	Precision	Recall	F1-score	Accuracy
Good Lighting	0.88	0.92	0.90	0.90
Low-lighting	0.76	0.74	0.75	0.80

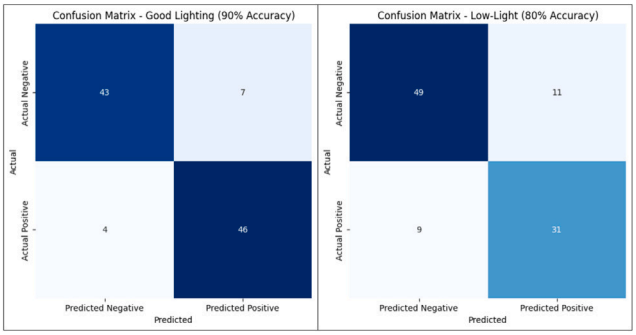


Fig. 4. Confusion matrix of the proposed system under good lighting and low lighting conditions.

Table 6 shows the performance comparison of the proposed system between low-vision and blind users. In Table 6 it is indicated that low vision users demonstrate significantly better ease of use (91% vs. 79%, $p = 0.001$) and satisfaction (4.4 vs. 4.0, $p = 0.003$), suggesting that blind users may benefit from additional training or simplified interfaces. The difference in object recognition accuracy between groups is modest but significant (91. 2% vs 88. 7%, $p = 0.042$), indicating that the system performs well in all levels of impairment. Face recognition shows no significant difference ($p = 0.068$), suggesting that this feature is equally accessible to both groups.

5.2. Green and blue button results

As discussed in the system design section, the system contains two buttons, a green button and a blue button. Pressing the green button prompts the system to ask the user to identify individuals in front of them through voice, storing this information for future reference by the core system.

When the blue button is pressed, the system captures an image, which is then processed by the LVLM to generate detailed information about the image and convey it to the user by voice assistance. This way, the user can obtain a detailed description of its surroundings and take better decisions. Figs. 8 and 9 indicate the system's blue button pressing results. This process is outlined in Figs. 5, 6, and 7. The complete process of green and blue pressing results have been outlined in Fig. 1 and subsequently explained in Section 3.1.

5.3. Distribution of false positives and false negatives

Fig. 10 presents the histogram plot showing the distribution of false positives and false negatives in object detection under different lighting conditions. It shows that under good lighting, the accuracy of object detection is 90%, which is reduced to 80% under low lighting. Furthermore, the number of errors per detection session in low lighting are detected more as compared to good lighting. Mean values of false positives and false negatives are less in good lighting as compared to low lighting.

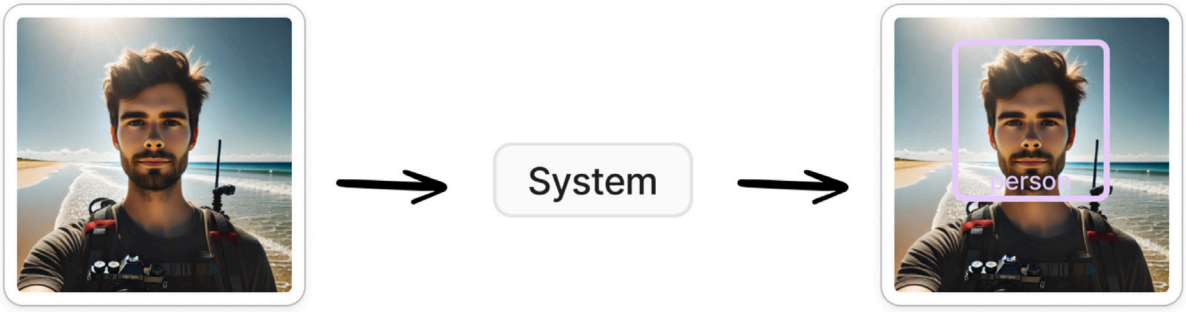


Fig. 5. Core system detecting image as shown in Fig. 1.

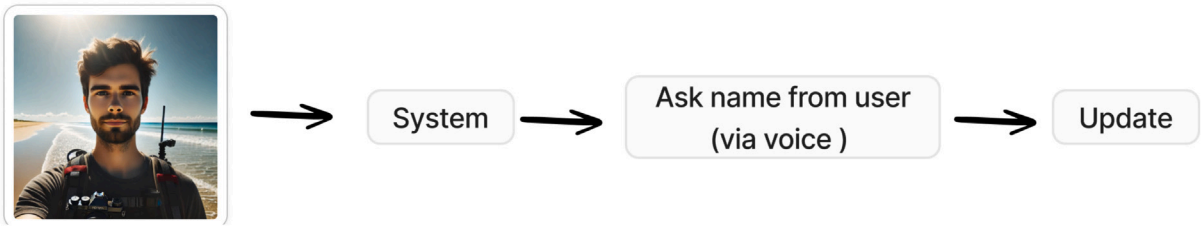


Fig. 6. Label updating process using green button as shown in Fig. 1.

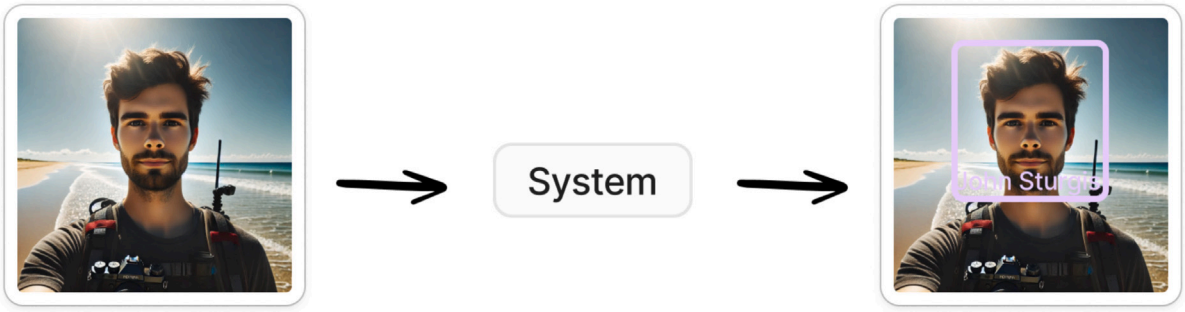


Fig. 7. Using core system again after updating the label.

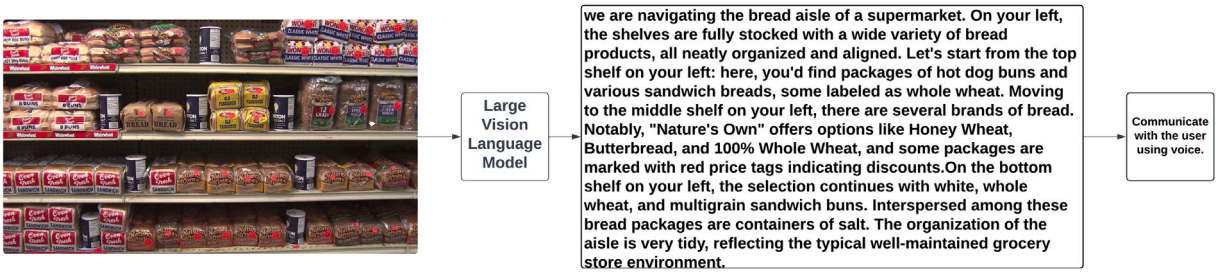


Fig. 8. Demo output of blue button (user is standing in a bread market).

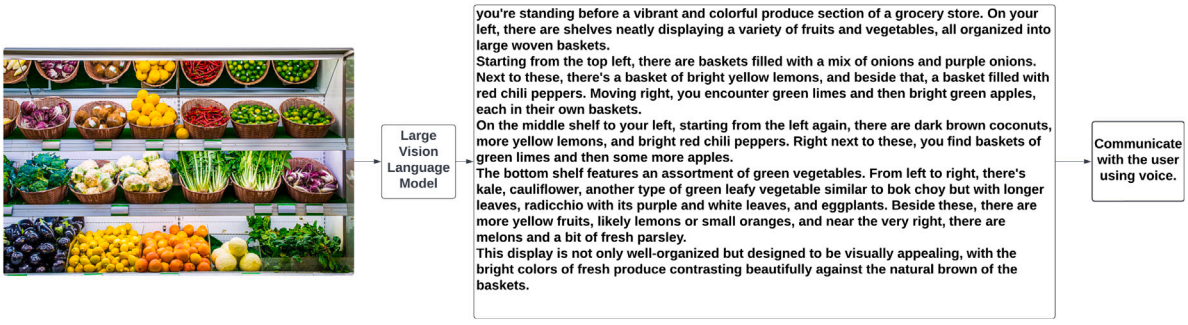


Fig. 9. Demo output of blue button (user is standing in a vegetable store).

Table 5
Object recognition by category (Good lighting).

Category	Precision	Recall	F1-score	Sample size	Key challenges
Furniture	0.85	0.88	0.86	150	Ambiguous shapes
Food Items	0.92	0.95	0.93	120	Similar textures
People/Faces	0.90	0.92	0.91	80	Occlusion, angles
Electronic Devices	0.89	0.91	0.90	100	Variable lighting
Text/Labels	0.88	0.90	0.89	95	Font variation
Navigation Obstacles	0.93	0.94	0.93	110	Critical safety
Overall Average	0.88	0.92	0.90	655	–

Table 6
System performance comparison between low vision and blind users.

Impairment level	Object Recognition	Face Recognition	Satisfaction Score (/5)	Ease of Use
Low Vision (N = 30)	91.2%	88.5%	4.4	91%
Blind (N = 20)	88.7%	85.2%	4.0	79%
		Statistical Significance		
p-value	$p = 0.042$*	$p = 0.068$	$p = 0.003$**	$p = 0.001$**

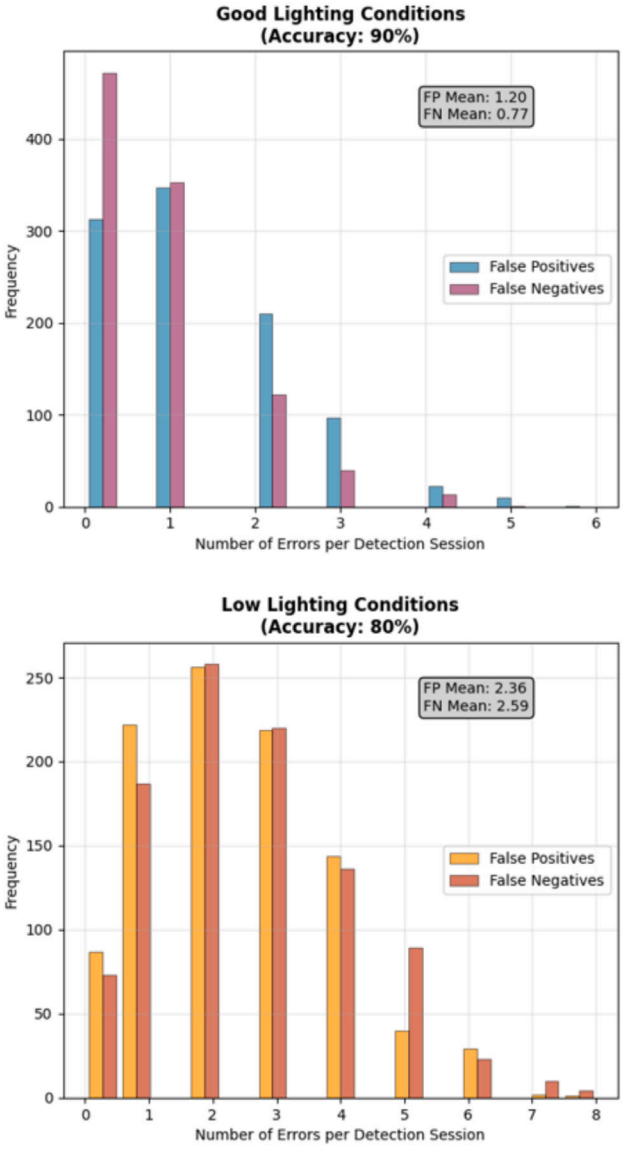


Fig. 10. Histogram plots showing the distribution of false positives and false negatives under different lighting conditions.

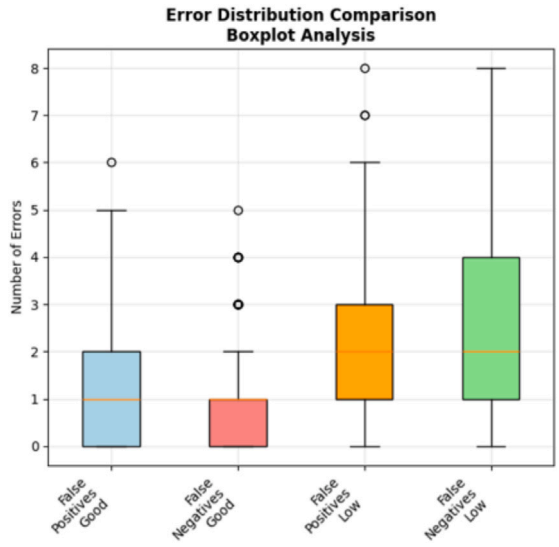


Fig. 11. Box plots showing the error variance of false positives and false negatives under different lighting conditions.

Fig. 11 presents the box plots showing the error variance of the object detection in good and low lighting conditions. It indicates that false negatives possess higher variability in low lighting as compared to good lighting. Furthermore, in good lighting, the error variance of the false positives is more as compared to low lighting conditions.

Category-specific error rates have been shown by the histogram presented in Fig. 12. This plot demonstrate that static objects like table, chair exhibit higher false positive rates, whereas, dynamic objects such as person, laptop show increased false negatives.

Violin plot analysis has been presented in Fig. 13. These shapes highlight the bimodal nature of false negatives in low lighting conditions, clearly indicating that in good lighting conditions, the error is much less as compared to low light conditions.

5.4. User satisfaction and usability

The system demonstrated exceptional user acceptance metrics during the usability evaluation. As shown in Table 7, the system's usability assessment yielded impressive average satisfaction and ease of use scores for the navigation, object identification, and shopping assistance functions of the system. On average 85% of participants found the



Fig. 12. Histogram plot showing the category-specific error rates.

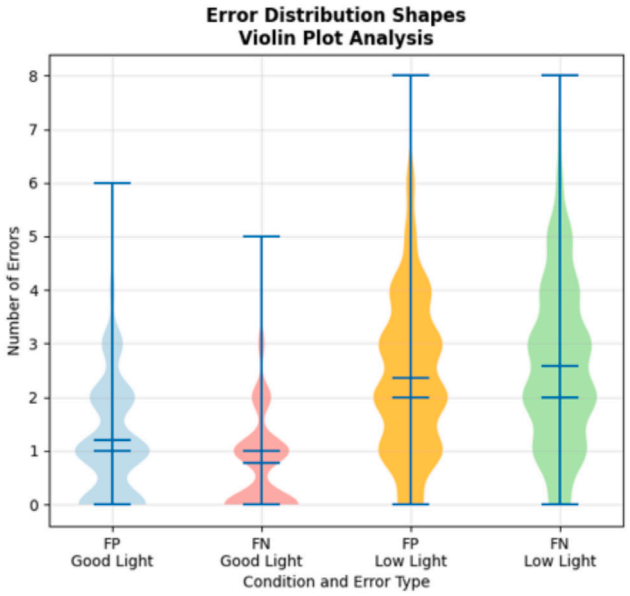


Fig. 13. Violin plot analysis of the error distribution in different lighting conditions.

device easy to use and on the scale of 5, on average 4.2 score was obtained about the users satisfaction from the device indicating that users felt quite satisfied after using the device.

Table 8 indicates that on average 85% users found the device very helpful in terms of different features of the device such that real time object detection, distance alert, and personalized data base showing their intention to incorporate it into their regular routines.

5.5. Response time

The system demonstrated efficient performance metrics across different operational modes. In real-time recognition scenarios, the system achieves a swift average response time of 1.5 s, enabling immediate object identification. For Detail Requests, which require more comprehensive processing through API calls, the system maintains an average response time of 5.5 s, with this additional duration primarily attributed to API call latency.

Table 7
Users' satisfaction scores.

Scenario	Average satisfaction score (/5)	Average ease of use score (%)	Suggestions
Navigation	4.5	90	Add more precise distance alerts
Object Identification	4.2	85	Improve recognition speed
Shopping Assistance	4.0	80	Better product categorization

Table 8
System's features usability.

Feature	Usage frequency (%)	Found very helpful (%)	Found less helpful (%)	User suggestions
Real-Time Object Detection	90	92	8	Add more object categories by default
Distance Alert	90	85	15	Increase warning range
Personalized Database	80	77	25	Simplify data entry process

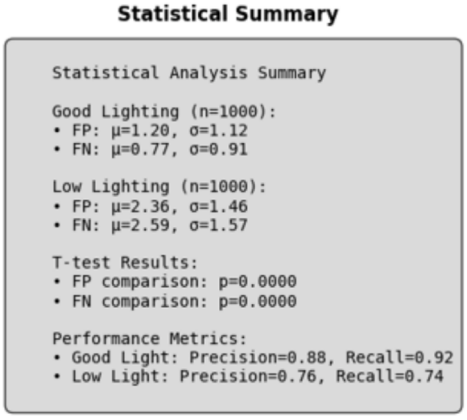


Fig. 14. Statistical summary of the object detection in different lighting conditions.

5.6. Statistical significance of results in good lighting

Fig. 14 presents the statistical summary of the object detection in different lighting conditions. Statistical analysis shows significant differences between lighting conditions for both error types ($p < 0.001$). The system demonstrates better precision (0.88) and recall (0.92) in good lighting as compared to low lighting scenarios where precision is 0.76 and recall is 0.74, suggesting conservative detection behavior.

In order to further judge the statistical significance of the achieved results in good lighting condition, a paired t-test was performed. As shown in Table 9 that paired t-tests confirmed recognition accuracy achieved by the proposed system (Mean = 90%, SD = 5%) compared to conventional devices (Mean = 63.4%, SD = 7%) with $p = 0.003$, 95% CI for difference [0.17, 0.32]. Similarly, improvements in average response time (Mean = 1.5 s vs 2.8 s) yielded $p = 0.001$, 95% CI for difference [−1.7, −0.8 s]. User satisfaction scores (Mean SUS score = 85 vs 70) also demonstrated significant improvement with $p = 0.004$, 95% CI for difference [7.1, 23.9].

5.7. Grad-CAM visualization of object detection

To better see how the MobileNetV2 SSD identifies main features in an image, we used Gradient-weighted Class Activation Mapping

Table 9
Statistical significance of the performance under good lighting conditions.

Metric	Our system	Baseline	p-value	95% CI
Recognition Accuracy (%)	90	63.4	0.003	[0.17, 0.32]
Response Time (s)	1.5	2.8	0.001	[−1.7, −0.8]
System Usability Score (out of 100)	85	70	0.004	[7.1, 23.9]



Fig. 15. Grad-CAM visualization highlighting the most discriminative regions (e.g., Face and Body) used by the MobileNetV2 SSD to classify the object as a Cat. The warmer colors (Yellow/Red) indicate high model attention, whereas cooler colors (Blue/Purple) indicate less relevance.

(Grad-CAM). Fig. 15 illustrates a sample Grad-CAM output where the MobileNetV2 SSD model detects a cat. It seems that the model relies mostly on the head and upper body of the cat to determine whether it is a cat or not. Such visualizations offer crucial insights into the decision-making process of DL models, enhancing trust and transparency, especially for assistive AI systems intended for safety-critical tasks. While Grad-CAM visualizations were used for the MobileNetV2 SSD model to illustrate attention during object detection, we did not apply Grad-CAM to FaceNet. This is because FaceNet is an embedding-based model that produces 128-dimensional feature vectors rather than class probabilities. Since it is not trained as a classifier, applying gradient-based visualization techniques like Grad-CAM is not straightforward without introducing additional proxy classification layers, which may compromise the purity of the face embedding process. Future work may explore attention mechanisms or surrogate visualizations specifically adapted for embedding-based models to enhance transparency in face recognition.

5.8. Training convergence curves

To validate the stability and convergence behavior of the MobileNetV2 SSD model fine-tuned on locally captured data, we plotted the training and validation loss and accuracy across 10 epochs. As illustrated in Fig. 16, the model exhibited rapid convergence within the first five epochs, followed by gradual stabilization. Both training and validation losses aligned closely without signs of overfitting. Training and validation accuracies also improved steadily, achieving final values of 93% and 92%, respectively. This behavior is typical of TL scenarios on small datasets and demonstrates the model's reliable learning dynamics on resource-constrained hardware.

5.9. Comparative analysis

The performance of the proposed system has been compared with the state-of-the-art commercially available systems in the literature such as AIris [69], MagicEye [70], NoorCam MyEye [20], DRISHTI [71], and NewVision [72]. Several features of the systems have been compared. Table 10 provides the comparative summary of the proposed system's features with the existing systems. Table 10 indicates that the proposed system incorporates several capabilities that are not simultaneously reported in the compared research prototypes. It provides several additional capabilities compared with previously reported systems, particularly in contextual understanding and personalization. In all such characteristics, the aforementioned conventional systems have been found deficient, while the proposed system, with the enhanced sense of environmental awareness, provides richer contextual descriptions. The system's capability to recognize acquaintances proved particularly valuable, leading to improved social comfort and more natural interactions.

Table 11 provides the comparative summary of the accuracies of the proposed system and the aforementioned systems. It is evident from Table 11 that the proposed system demonstrates competitive performance compared to previously reported research prototypes in terms of object detection and scene description performance. Furthermore, it may be noted from Table 11 that all the features provided by the proposed system are not available in the available systems altogether.

6. System's practical applications and extensions

The proposed system has strong potential for broader integration within emerging smart city infrastructures and personalized healthcare technologies.

Integration with Smart Cities and IoT Networks: The modular design of the system makes it compatible with future extensions such as the system can be connected later to smart traffic signals, navigation beacons based on Bluetooth Low Energy, or GPS-equipped public transport hubs. For instance, the system uses the information sent by IoT-connected pedestrian crossings to let users when it is safe for them to cross. Through APIs supplied by the smart city network, it is able to provide real-time bus arrival information or check the user's approximate position near stations, which improves how the app operates autonomously. With the use of edge AI accelerators such as Jetson Nano, the integration process runs quickly and with full understanding of the surrounding events.

Scalability and Personalization for Diverse Users: The system supports incremental personalization by allowing users to add new objects or faces via a voice-controlled workflow. This makes it adaptable to users with varying degrees of visual impairment, from low vision to complete blindness. For instance, users with partial sight may use it as a supplementary system for fine-grained contextual understanding (e.g., identifying shopping items), while blind users may rely more heavily on its obstacle detection, facial recognition, and auditory navigation. In future versions, a user profile management system can allow each user to customize feedback modes (e.g., vibration, audio pitch), language, or sensitivity thresholds based on their unique needs.

Healthcare and Community Use Cases: On top of home use, the system can benefit public institutions (for example, hospitals, libraries) because it allows multiple users to use the same system at the same time. Wearable biometric devices can be used in healthcare to respond by alerting the user if they become too stressed or their heart rate goes above usual levels while getting directions.

As a result of these capabilities, the system can be used both on its own and as an important piece of inclusive smart city technology.

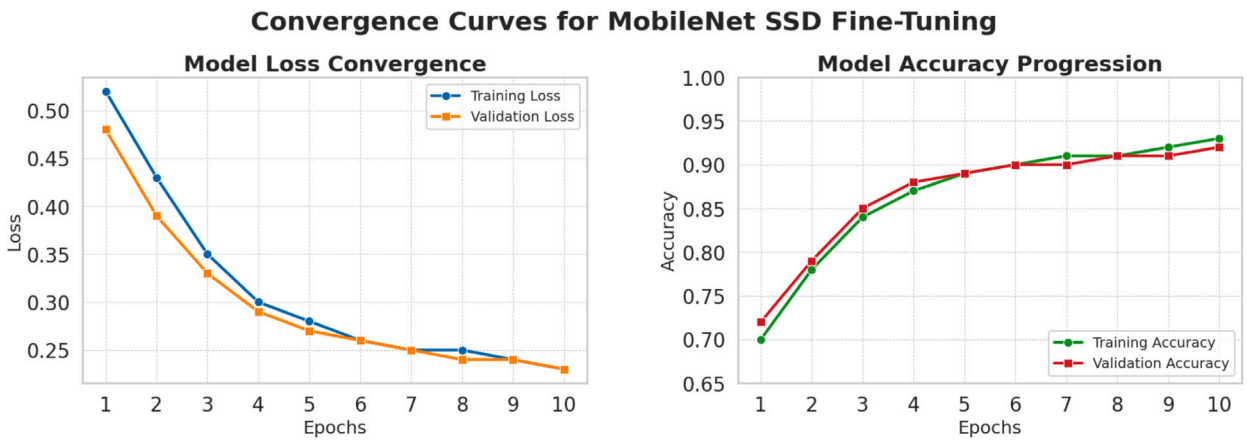


Fig. 16. Training and validation loss and accuracy convergence curves for the fine-tuned MobileNetV2 SSD model. The model rapidly converges within the first five epochs, with both loss values stabilizing and accuracy metrics achieving above 90%. This confirms effective TL behavior on the local dataset.

Table 10
Feature-level comparison with commercial systems.

Features	Proposed system	AIris [69]	MagicEye [70]	NoorCam MyEye [20]	DRISHTI [71]	NewVision [72]
Object Detection	Real-time object recognition with LVLM for enhanced accuracy and contextual insights	Basic object detection for navigation and task assistance	CNN-based model for general object recognition	Recognizes objects via barcodes; less flexible for real-time detection	Detects objects and obstacles for navigation	Utilizes DL for object identification and obstacle avoidance
Contextual Understanding	Deep contextual understanding using large vision-language models (LVLM)	Scene description provided but lacks in-depth contextual analysis	Limited contextual understanding of object interactions	Provides audio descriptions without comprehensive contextual understanding	Minimal contextual insights; primarily focused on navigation	Integrates voice recognition and assistants for real-time environmental information
Face Recognition	Personalized face recognition with a user-updated database along with real-time database updation	Limited facial recognition for basic identification tasks	Includes facial recognition	Strong facial recognition capability	Does not support face recognition	Capable of identifying people through facial recognition
Navigation Assistance	Integrated obstacle detection and personalized audio feedback	Basic navigation support with distance alerts	Provides proximity alerts but lacks advanced navigation feedback	Not designed for navigation	Core functionality with obstacle detection and auditory alerts	Assists in navigation using a combination of sensors and voice commands
Personalization	Fully user-centric with customizable object and face databases	General-purpose; no personalization options	No database for personalization	No user-updated database; relies on pre-trained recognition data	Limited personalization; navigation is generic	Allows interaction through voice commands for personalized assistance

Table 11
Accuracy comparison of the proposed system with the state-of-the-art.

Feature	Proposed system	AIris [69]	MagicEye [70]	NoorCam MyEye [20]	DRISHTI [71]	NewVision [72]
Object Detection	80% accuracy in low lighting and 90% accuracy in good lighting.	63.4% mean Average Precision on COCO dataset% accuracy for general objects	NA	NA	NA	NA
Face Recognition	84.77% accuracy in low lighting and 87.22% accuracy in good lightning on LFW dataset [63]	99.38% accuracy on LFW dataset	94.5% accuracy on LFW dataset	NA	NA	NA
Scene Description	88% BLEU score on Flickr8k dataset [64]	27% BLEU score on Flickr8k dataset	NA	NA	NA	NA

7. Challenges and limitations

Current implementation faces several technical constraints, such as limited processing and latency, that warrant consideration. Although the Raspberry Pi proves capable for basic operations, it imposes limitations on the complexity of models that can be executed locally, potentially restricting more advanced processing capabilities, especially when executing real-time computer vision inference tasks. The computational limitations of the Raspberry Pi restrict the use of heavier AI models and larger neural networks. Despite optimizations such as model quantization using TensorFlow Lite and parallel threading of processes, achieving sub-second inference time for complex models remained difficult. It took about 1.2 s for a single frame on the SSD model. The total time needed for real-time object and face recognition was 1.5 s, as including face recognition with FaceNet increased the process by 0.3 s. The procedure that involves obtaining contextual detail

requests, making an API call to GPT-4o mini with Azure, increased the time it took to respond to 5.5 s, mostly due to network delay and the processing of the cloud model. The system also faces thermal throttling during continuous extended use, affecting processing consistency. Environmental factors also impact system performance, particularly in low-light conditions. Additionally, the system's reliance on network connectivity for certain features presents a notable dependency that could affect functionality in areas with limited internet access.

8. Future directions and improvements

To evolve the proposed system into a more robust, intelligent, and user-friendly platform, several future improvements and directions are envisioned:

Speech Recognition and Gesture-Based Interaction: To enhance hands-free usability, future iterations may integrate speech recognition for system control (e.g., saying "what is in front of me?" instead of pressing a button). Open-source tools like Vosk or Whisper could be deployed locally for privacy. Additionally, incorporating simple gesture-based interactions via an IMU (Inertial Measurement Unit) or vision-based hand tracking can offer intuitive ways to trigger commands, especially useful for users with mobility impairments or who cannot access physical buttons.

Edge AI Deployment: While this version uses Raspberry Pi 4 for local processing and offloads contextual description to GPT-4o mini via Azure, future versions may transition to more capable edge AI hardware such as NVIDIA Jetson Nano, Jetson Xavier NX, or Google's Coral TPU. This would allow running larger, more accurate models entirely on-device, thereby reducing cloud dependency, improving privacy, and decreasing latency.

Multimodal Sensor Fusion: To enhance object and obstacle detection across varied lighting and environmental conditions, the system can be augmented with additional sensing modalities such as:

LiDAR: For precise 3D depth mapping and obstacle positioning.

Infrared (IR) Cameras: For low-light or nighttime operation.

Thermal Sensors: Useful in detecting living beings or navigating through fog/smoke. Fusing these sensors with RGB camera input using Kalman filters or deep sensor fusion networks could enable a more robust and reliable scene understanding.

AI-Driven Adaptive Learning: Personalization can be advanced through AI-driven adaptive learning techniques. The system can track user preferences (e.g., commonly accessed items, frequent locations, familiar faces) and adjust its behavior over time. Reinforcement learning or continual learning frameworks could be used to adapt model responses, prioritize certain objects, and refine audio feedback to better match individual user needs.

Real-Time Multilingual Support: To support users across different regions, the -speech engine and GPT-based description module can be adapted to provide feedback in multiple languages. This would significantly enhance accessibility for non-English speakers in global deployments.

These directions aim to move the system from a reactive tool to a proactive, context-aware companion adaptable to diverse user needs and smart city ecosystems.

9. Conclusion

This study presented an AI-based wearable vision assistance system integrating edge-based object detection, incremental personalized recognition, and LVLM-driven contextual scene understanding. The system operates on affordable hardware using a Raspberry Pi platform and achieved 90% object recognition accuracy under good lighting conditions along with high usability scores (85%) in a study involving 50 visually impaired participants. The results demonstrate the feasibility of combining lightweight computer vision models with large vision-language models to provide both recognition and contextual awareness in assistive devices. The proposed architecture highlights the potential for cost-effective, customizable, and context-aware assistive technologies for visually impaired users.

CRediT authorship contribution statement

Mirza Samad Ahmed Baig: Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Syeda Anshrah Gillani:** Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Shahid Munir Shah:** Writing – review & editing, Visualization, Validation, Supervision, Project administration, Methodology, Investigation, Formal analysis. **Mahmoud Aljawarneh:** Resources, Project administration, Funding acquisition. **Abdul Akbar Khan:** Formal analysis, Conceptualization. **Muhammad Hamzah Siddiqui:** Visualization, Resources.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author(s) used ChatGPT in order to paraphrase sentences. After using this tool/service, the author (s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

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Declaration of competing interest

The authors certify that they have NO affiliations with or involvement in any organization or entity with any financial interest (such as honoraria; educational grants; participation in speakers' bureaus; membership, employment, consultancies, stock ownership, or other equity interest; and expert testimony or patent-licensing arrangements), or non-financial interest (such as personal or professional relationships, affiliations, knowledge or beliefs) in the subject matter or materials discussed in this manuscript.

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Data availability

Data will be made available on request.

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